

# Statistical Analysis of Sepsis-3 in Critically Ill Patients (MIMIC-IV)

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01 July 2026

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# 1 Introduction

Sepsis is life-threatening organ dysfunction caused by a dysregulated host response to infection, and it remains one of the leading causes of death in intensive care units worldwide. The **Sepsis-3** consensus (Singer et al., 2016) defines sepsis operationally as a **suspected infection** accompanied by an acute rise of **two or more points** in the Sequential Organ Failure Assessment (**SOFA**) score.

This report analyses **MIMIC-IV v3.1**, a large, de-identified database of patients admitted to the critical-care units of the Beth Israel Deaconess Medical Center. From the raw tables a reproducible pipeline (scripts 01–05) builds an adult, first-ICU-stay cohort, detects suspected infection from antibiotic orders paired with cultures, computes the six SOFA organ scores from the first 24 hours of each stay, and applies the Sepsis-3 rule. The result is an analysis matrix of **65,241** ICU stays, of which **37.1%** meet the Sepsis-3 definition.

The clinical question is whether **routinely measured physiology that is *not* part of the sepsis definition** — lactate, white-cell count, heart rate, respiratory rate, temperature, and age — can separate septic from non-septic stays, and what different statistical methods reveal about the same data. Choosing predictors outside the SOFA score avoids circularity: the models are asked to recover the label from independent signals, not from the very quantities used to construct it.

All inferential and modelling sections use the **36,529** stays with a complete set of these six measurements (50.0% Sepsis-3 positive). Because lactate is ordered preferentially in sicker patients, this complete-case subset is enriched for severity relative to the full cohort; this is revisited as a limitation in the conclusion.

The report follows a structured progression:

1. **Exploratory data analysis** (Section 2) describes the cohort and feature distributions by sepsis status.
2. **Hypothesis testing** (Section 3) quantifies which features differ between groups, focusing on effect size.
3. **Maximum likelihood estimation** (Section 4) models the typical lactate of a septic patient.
4. **Generalised linear models** (Section 5) determine which features carry independent predictive power.
5. **Kernel density estimation and mixture modelling** (Section 6) attempt to recover the two groups without labels.
6. **Metropolis-Hastings sampling** (Section 7.1) re-expresses the likelihood result in a Bayesian framework.
7. **A Hidden Markov Model** (Section 7.2) tests whether the ordering of observations carries extra structure.
8. **Predictive validation** (Section 8) subjects the model to out-of-sample cross-validation, calibration, precision–recall and decision-curve analysis, a non-linearity check, and a formal treatment of missing data.

## 2 Exploratory Data Analysis

The aim of this section is to understand the data before modelling. The focus is on lactate, white-cell count, and the vital signs, which are central to the recognition of sepsis at the bedside.

In the modelling subset there are **18,255** non-septic and **18,274** septic stays (50.0% septic). The classes are imbalanced at roughly **1.0:1**, milder than many clinical-outcome datasets and not severe enough to require resampling; the inferential focus on effect size further reduces its impact.

Table 1: Summary statistics for the six features, by sepsis status.

| Feature                | Mean (no) | Mean (sep) | Med. (no) | Med. (sep) | SD (no) | SD (sep) |
|------------------------|-----------|------------|-----------|------------|---------|----------|
| Age (years)            | 64.17     | 65.38      | 66.00     | 67.00      | 16.67   | 16.40    |
| Heart rate, max (bpm)  | 102.59    | 108.89     | 99.00     | 106.00     | 76.68   | 50.90    |
| Lactate, max (mmol/L)  | 2.58      | 3.10       | 2.10      | 2.20       | 2.26    | 2.87     |
| Resp. rate, max (/min) | 28.30     | 541.79     | 27.00     | 28.00      | 31.66   | 54637.16 |
| Temperature, max (C)   | 37.25     | 37.56      | 37.17     | 37.39      | 0.74    | 0.97     |
| WBC, max (K/uL)        | 14.05     | 16.06      | 12.80     | 14.20      | 9.32    | 13.59    |

Table 1 shows that septic stays tend to have higher lactate, higher white-cell counts, and faster heart and respiratory rates. These shifts match the bedside picture of sepsis, although the spread within each group is wide.

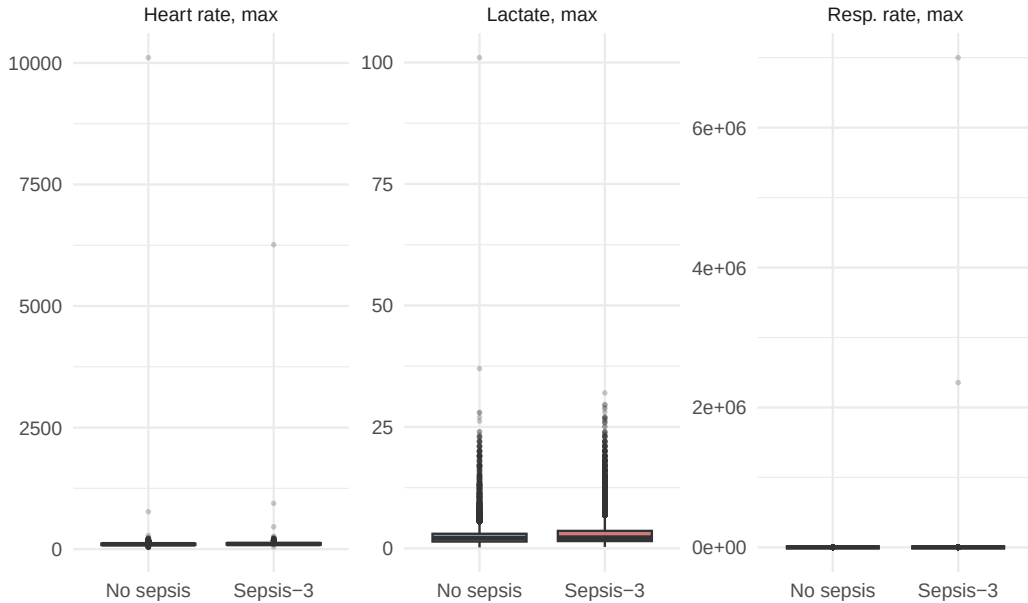


Figure 1: Three features by sepsis status. Septic stays show higher lactate, faster breathing, and faster heart rates, with substantial overlap.

Figure 1 confirms the direction of these differences but also the overlap: no single feature cleanly separates the groups, which motivates the multivariable model in Section 5.

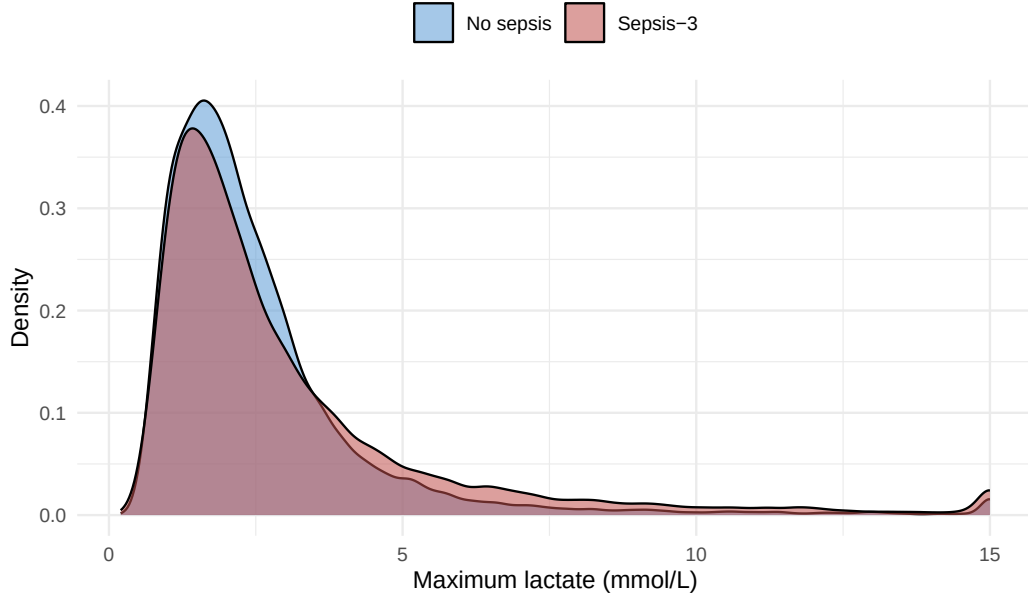


Figure 2: Distribution of maximum lactate by sepsis status (values capped at 15 mmol/L for display). Both groups are right-skewed; the septic distribution is shifted upward.

Figure 2 shows two overlapping, right-skewed distributions. The septic curve is shifted toward higher lactate, but the long right tail in both groups signals that a log transform will be useful for the distributional models in Sections 4–7.2.

*Collectively, this section shows the data contain genuine, clinically coherent signal, but no single feature is sufficient for perfect separation.*

### 3 Hypothesis Testing

This section compares septic and non-septic stays on three features using three different tests. With **36,529** observations, even trivial differences yield very small p-values, so interpretation focuses on **effect size** — the magnitude of the difference — rather than significance alone.

#### 3.1 Test 1: Maximum lactate (Welch t-test)

A Welch t-test is used because the two groups have unequal variances.

$H_0$ : mean maximum lactate is equal in septic and non-septic stays

$H_1$ : mean maximum lactate differs between the groups

Table 2: Welch t-test: maximum lactate by sepsis status.

| Statistic        | Value          |
|------------------|----------------|
| t                | -19.18         |
| df               | 34642          |
| p-value          | <1e-16         |
| Mean (no sepsis) | 2.58           |
| Mean (Sepsis-3)  | 3.10           |
| Mean difference  | -0.52          |
| Cohen's d        | -0.20          |
| 95% CI for d     | [-0.22, -0.18] |

Table 2 shows mean maximum lactate of **2.58** mmol/L in non-septic stays versus **3.10** mmol/L in septic stays, a difference of **-0.52** mmol/L. The p-value is <1e-16, so the difference is not attributable to chance. Cohen's *d* of **-0.20** indicates a small-to-moderate effect: lactate separates the groups, but not perfectly.

### 3.2 Test 2: Maximum white-cell count (Wilcoxon rank-sum)

White-cell counts are heavily right-skewed, so a non-parametric Wilcoxon rank-sum test is used.

$H_0$ : the distribution of maximum WBC is the same in both groups

$H_1$ : the distributions differ

Table 3: Wilcoxon rank-sum: maximum WBC by sepsis status.

| Statistic       | Value          |
|-----------------|----------------|
| W               | 148,333,588    |
| p-value         | <1e-16         |
| Rank-biserial r | -0.11          |
| 95% CI          | [-0.12, -0.10] |

Table 3 shows a significant difference in white-cell counts. The rank-biserial correlation of **-0.11** is small in magnitude — weaker than lactate, consistent with white-cell count being a noisier marker that rises in many non-septic stresses as well.

### 3.3 Test 3: Tachypnoea and sepsis (Chi-squared)

Respiratory rate is dichotomised at the **qSOFA** threshold of 22 breaths/min (a recognised marker of clinical deterioration), and its association with sepsis is tested with a chi-squared test of independence.

$H_0$ : tachypnoea ( $\geq 22$ /min) and sepsis are independent

$H_1$ : they are associated

Table 4: Chi-squared: tachypnoea (RR  $\geq 22$ ) by sepsis status.

| Statistic   | Value        |
|-------------|--------------|
| Chi-squared | 176.2        |
| df          | 1            |
| p-value     | $<1e-16$     |
| Cramer’s V  | 0.07         |
| 95% CI      | [0.06, 1.00] |

Table 4 shows a significant association between tachypnoea and sepsis, with Cramer’s V of **0.07** (a weak association). Dichotomising at a single threshold discards information; the GLM in Section 5 uses the continuous respiratory rate and recovers a stronger signal.

*Collectively, all three features carry genuine signal. Lactate shows the largest effect; white-cell count and tachypnoea are weaker but real.*

## 4 Maximum Likelihood Estimation

This section estimates the typical lactate of a septic patient. Because lactate is strictly positive and right-skewed, it is modelled as **log-normal**: the log of maximum lactate is assumed Normal. Maximum likelihood then estimates the mean  $\mu$  and standard deviation  $\sigma$  on the log scale, and the typical (median) lactate is recovered as  $e^\mu$ .

Table 5: MLE for the log-normal model of septic maximum lactate.

| Parameter                           | Estimate     |
|-------------------------------------|--------------|
| mu (mean log-lactate   sepsis)      | 0.865        |
| sigma (SD log-lactate   sepsis)     | 0.684        |
| n (septic)                          | 18,274       |
| SE of mu                            | 0.0051       |
| Median lactate $\exp(\mu)$ (mmol/L) | 2.37         |
| 95% CI for median                   | [2.35, 2.40] |

The maximum-likelihood estimate from Table 5 shows:

1. The typical (median) maximum lactate for a septic stay is about **2.37** mmol/L — above the 2 mmol/L threshold used in the septic-shock definition.
2. The log-scale standard deviation of **0.684** reflects the wide spread of lactate values.
3. The very small standard error shows the central value is estimated precisely, a direct consequence of the large septic sample.

## 5 Generalised Linear Models

Logistic regression is used to model the probability of sepsis from several features simultaneously, because the outcome is binary. Two models are compared:

- **Model 1** uses maximum lactate only.

- **Model 2** adds white-cell count, heart rate, respiratory rate, temperature, and age.

A correlation check among the continuous predictors shows the strongest pairwise correlation is **0.06**, well below the 0.8 threshold that would signal redundancy, so all predictors are retained.

## 5.1 Logistic Regression Coefficients

Table 6: Logistic regression coefficients (log-odds scale).

| Model            | Term          | Estimate | SE    | p-value |
|------------------|---------------|----------|-------|---------|
| M1: lactate only | (Intercept)   | -0.238   | 0.016 | <1e-16  |
| M1: lactate only | lactate_max   | 0.085    | 0.005 | <1e-16  |
| M2: 6 predictors | (Intercept)   | -21.578  | 0.604 | <1e-16  |
| M2: 6 predictors | lactate_max   | 0.078    | 0.005 | <1e-16  |
| M2: 6 predictors | wbc_max       | 0.018    | 0.001 | <1e-16  |
| M2: 6 predictors | hr_max        | 0.008    | 0.001 | <1e-16  |
| M2: 6 predictors | resp_rate_max | 0.003    | 0.001 | 0.005   |
| M2: 6 predictors | temp_max_c    | 0.522    | 0.016 | <1e-16  |
| M2: 6 predictors | age           | 0.011    | 0.001 | <1e-16  |

Coefficients in Table 6 are on the log-odds scale. Converting to odds ratios aids interpretation.

Table 7: Model 2 odds ratios (per one-unit increase).

| Term          | Odds ratio | 95% CI         |
|---------------|------------|----------------|
| lactate_max   | 1.081      | [1.071, 1.091] |
| wbc_max       | 1.018      | [1.015, 1.021] |
| hr_max        | 1.008      | [1.007, 1.009] |
| resp_rate_max | 1.003      | [1.001, 1.006] |
| temp_max_c    | 1.685      | [1.633, 1.739] |
| age           | 1.011      | [1.009, 1.012] |

The odds ratio for maximum lactate is **1.08**, meaning each additional 1 mmol/L multiplies the odds of sepsis by about 1.08, holding the other variables constant. The vital signs and age each contribute smaller but independent effects, with odds ratios above 1 indicating that faster heart and respiratory rates, higher temperature, and older age are each associated with a higher probability of sepsis once the others are accounted for.

## 5.2 Model Comparison

Table 8: Model comparison (lower AIC is better).

| Model              | AIC     | Residual deviance | df    |
|--------------------|---------|-------------------|-------|
| M1: lactate only   | 50257.1 | 50253.1           | 36527 |
| M2: six predictors | 48112.3 | 48098.3           | 36522 |

Table 8 shows the AIC falling from **50257** (lactate only) to **48112** (six predictors), a clear improvement: the additional features carry information about sepsis beyond lactate alone. The six-predictor model achieves an approximate AUC of **0.66**, indicating modest discrimination from physiology that is external to the sepsis definition.

*Sepsis is therefore best explained by a combination of features rather than any single measurement.*

## 6 Density Estimation and Mixture Models

The sepsis labels are now ignored to ask whether maximum lactate alone splits the data into two natural groups.

### 6.1 Kernel Density Estimate

Silverman’s rule-of-thumb bandwidth is

$$h = 0.9 \times \min\left(s, \frac{IQR}{1.34}\right) \times n^{-1/5},$$

where  $s$  is the standard deviation,  $IQR$  the interquartile range, and  $n$  the sample size. Because lactate is right-skewed, the KDE is computed on the **log** scale, where the distribution is closer to symmetric.

The computed bandwidth is **0.068** on the log scale.

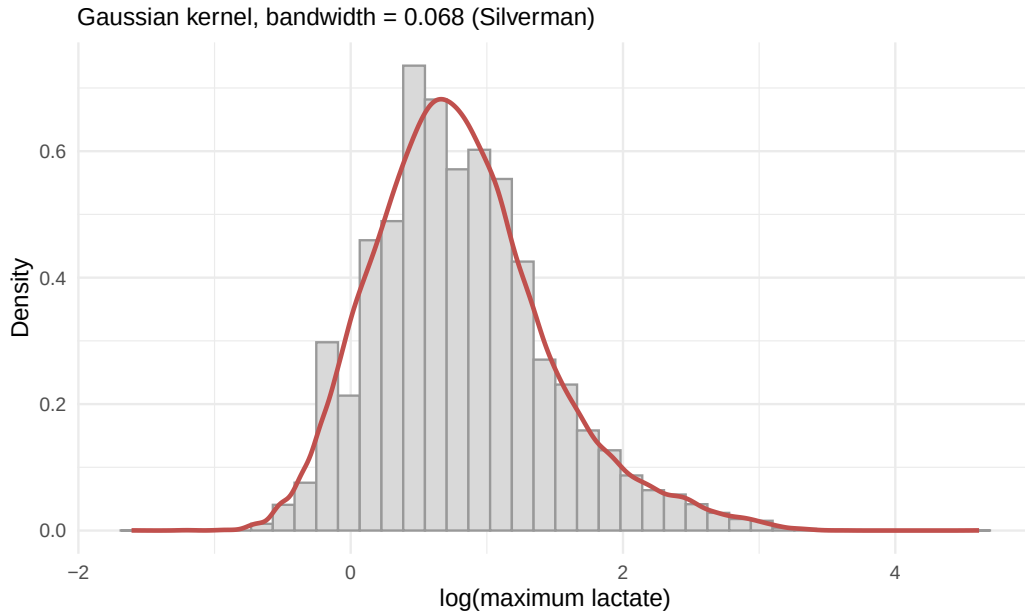


Figure 3: Kernel density estimate of log maximum lactate (Gaussian kernel, Silverman bandwidth). A shoulder above the main peak hints at a second, higher-lactate population.

Figure 3 shows a dominant peak with a right shoulder, suggesting the data may be a blend of a larger normal-lactate population and a smaller high-lactate one rather than a single homogeneous group.

## 6.2 Two-component Gaussian mixture

A two-component Gaussian mixture is fitted to log-lactate by the Expectation-Maximization algorithm, using no labels:

$$f(x) = \pi_1 \mathcal{N}(\mu_1, \sigma_1^2) + (1 - \pi_1) \mathcal{N}(\mu_2, \sigma_2^2).$$

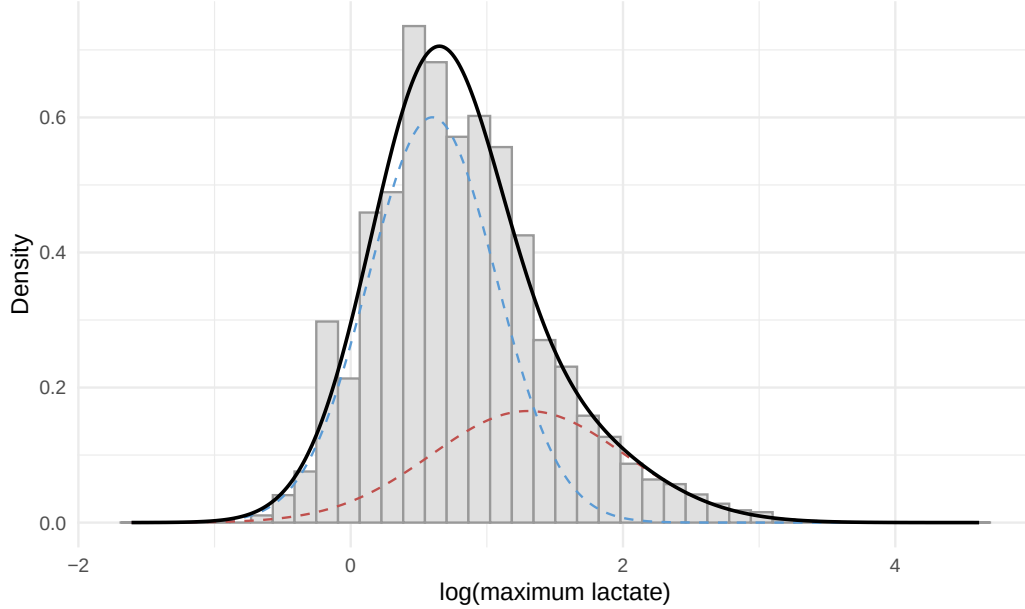


Figure 4: Two-component Gaussian mixture (dashed components, solid sum) fitted to log maximum lactate.

The mixture places one component near **1.82** mmol/L (back-transformed) and the other near **3.68** mmol/L, with mixing weights 0.70 and 0.30.

Matching the unlabelled components to the true labels gives an unsupervised agreement of **54.3%**. Unlike a marker such as tumour radius — which nearly separates its two classes — lactate alone only **partially** recovers sepsis status: the two populations overlap heavily, so a single biomarker is a weak unsupervised classifier. This is itself an informative negative result — it shows why the Sepsis-3 definition combines infection with a multi-organ score rather than relying on any one number.

## 7 Bayesian Inference and MCMC

Maximum likelihood (Section 4) gave a single best estimate of the mean log-lactate. Bayesian inference instead returns a full posterior distribution, expressing uncertainty as a range of plausible values.

### 7.1 Part A: Bayesian inference via Metropolis-Hastings

#### 7.1.1 Model setup

Log maximum lactate for septic stays is modelled as Normal with unknown mean  $\mu$  and a known standard deviation fixed at the MLE  $\hat{\sigma}$  from Section 4. A weak prior  $\mu \sim \mathcal{N}(\log 2, 1^2)$  is centred at a lactate of 2 mmol/L — the clinical threshold — but is wide enough to let the data dominate.

### 7.1.2 Likelihood and posterior

The posterior combines the Normal likelihood with the Normal prior,

$$p(\mu | x) \propto p(x | \mu) p(\mu),$$

which here is conjugate (and therefore Normal). The Metropolis-Hastings algorithm is nonetheless used to demonstrate a general sampling approach that does not rely on conjugacy.

Two chains are run from deliberately poor, opposite starting points and the first 2,000 iterations discarded as burn-in.

### 7.1.3 Convergence diagnostics

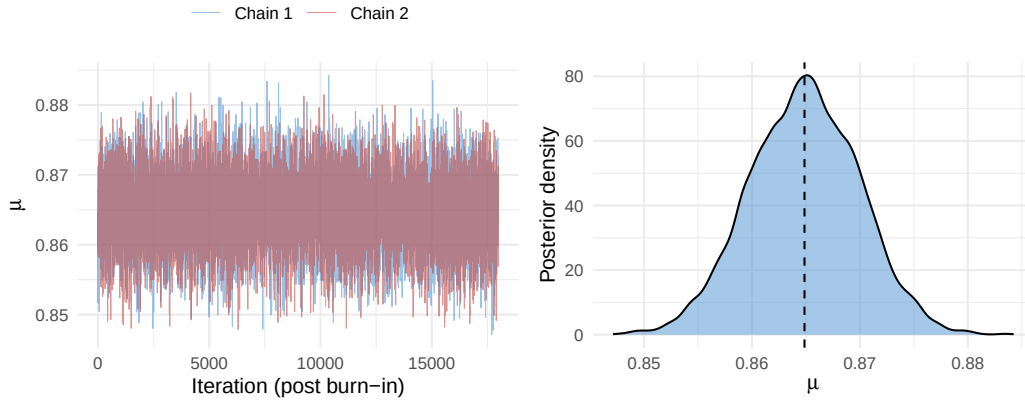


Figure 5: Trace plots (left) of two chains overlap after burn-in; posterior density (right) with the MLE marked (dashed).

The chains overlap after burn-in (Figure 5). The Gelman-Rubin statistic is  $\hat{R} = 1.000$  (values below 1.05 indicate convergence) and the effective sample size is about **4,131**, so the posterior samples are reliable.

### 7.1.4 Posterior summary and comparison with MLE

Table 9: MLE versus Bayesian posterior for mean log-lactate.

| Quantity                        | MLE            | Bayes (M-H)    |
|---------------------------------|----------------|----------------|
| Mean log-lactate   sepsis       | 0.865          | 0.865          |
| SE / posterior SD               | 0.0051         | 0.0051         |
| 95% interval                    | [0.855, 0.875] | [0.855, 0.875] |
| Implied median lactate (mmol/L) | 2.37           | 2.37           |

Table 9 shows the Bayesian and maximum-likelihood estimates are almost identical. With a large septic sample and a weak prior, the data dominate; a much stronger prior centred at log 2 would be needed to pull the estimate appreciably toward 2 mmol/L.

*The two frameworks agree, but the Bayesian result additionally provides a full posterior — more valuable when samples are small or strong prior knowledge exists.*

## 7.2 Part B: Hidden Markov Model

The mixture model treated stays as independent. A hidden Markov model allows consecutive stays to share a latent state via a transition matrix.

**Note on ordering.** The stays are not a true time series; their order is the processing order. The question here is therefore deliberately narrow: *after splitting the data into two lactate states, does the ordering add any structure?* If not, the HMM should reproduce the mixture model — and this negative result is itself informative. The HMM’s natural home is the *within-patient* hourly vital-sign sequence, where a latent state genuinely evolves toward organ dysfunction; that temporal use is the subject of the follow-on longitudinal analysis (see the Conclusion) and is outside the scope of this cross-sectional report.

### 7.2.1 Choosing the number of states

Table 10: HMM model selection by BIC (lower is better).

| States | Log-likelihood | BIC     |
|--------|----------------|---------|
| 1      | -7835.3        | 15688.5 |
| 2      | -7554.1        | 15171.1 |
| 3      | -7544.5        | 15214.9 |

Table 10 selects the **2-state** model by BIC, consistent with the two-population structure seen in Section 6.

### 7.2.2 Emissions and transition matrix

Table 11: HMM emissions versus the Section 6 mixture components.

| State            | Emission mean (log) | Median lactate (mmol/L) | Mixture mean (Sec. 6) |
|------------------|---------------------|-------------------------|-----------------------|
| State 1 (lower)  | 0.649               | 1.914                   | 0.600                 |
| State 2 (higher) | 1.629               | 5.096                   | 1.303                 |

Table 12: Estimated transition matrix.

| From    | To State 1 | To State 2 |
|---------|------------|------------|
| State 1 | 0.822      | 0.178      |
| State 2 | 0.818      | 0.182      |

Table 11 shows the HMM emission means almost coincide with the mixture components from Section 6.2: both methods place the two lactate populations in the same locations. The transition matrix (Table 12) has a stationary distribution of (0.82, 0.18). Any “stickiness” on the diagonal is the pattern EM tends to produce when fitting independent, overlapping data, not evidence of real sequential dependence.

### 7.2.3 Unsupervised classification accuracy

Decoding the most likely state for each stay and matching states to labels gives an unsupervised accuracy of **53.8%** — essentially the same as the mixture model (54.3%). The HMM learns nothing beyond the mixture: the two-lactate-population structure is a property of the **patients**, not of the order in which they were recorded.

## 8 Predictive Validation, Calibration and Missingness

The GLM in Section 5 reported an AUC computed **on the same data it was fitted to**, which is optimistically biased. A model intended — ultimately — to support *real-time* prediction must be judged on data it has not seen, must be **calibrated** (predicted risks must match observed frequencies), and must be honest about the **missing-data** mechanism that shaped the analysis sample. This section addresses all three.

### 8.1 Out-of-sample discrimination

Because the cohort keeps only each patient’s first ICU stay, every patient contributes exactly one row, so an ordinary random split is already patient-disjoint (no leakage between train and test). Ten-fold cross-validation is used: each stay is scored by a model trained on the other nine folds.

Table 13: In-sample versus cross-validated discrimination.

| Model              | In-sample AUC | 10-fold CV AUC | 95% CI (CV, M2) |
|--------------------|---------------|----------------|-----------------|
| M1: lactate only   | 0.540         | 0.539          |                 |
| M2: six predictors | 0.661         | 0.661          | [0.655, 0.666]  |

Table 13 shows the cross-validated AUC of the six-predictor model is **0.661** (95% CI 0.655<U+2013>0.666), essentially unchanged from the in-sample **0.661** — with 36,529 observations and only six predictors the model does not overfit. A DeLong test confirms the six-predictor model discriminates better than lactate alone (AUC 0.661 vs 0.539,  $p < 1e-16$ ).

### 8.2 Calibration and clinical usefulness

Discrimination (ranking) is not the same as calibration (the numerical risk being correct). The cross-validated risks are grouped into deciles and the mean predicted risk compared with the observed sepsis rate.

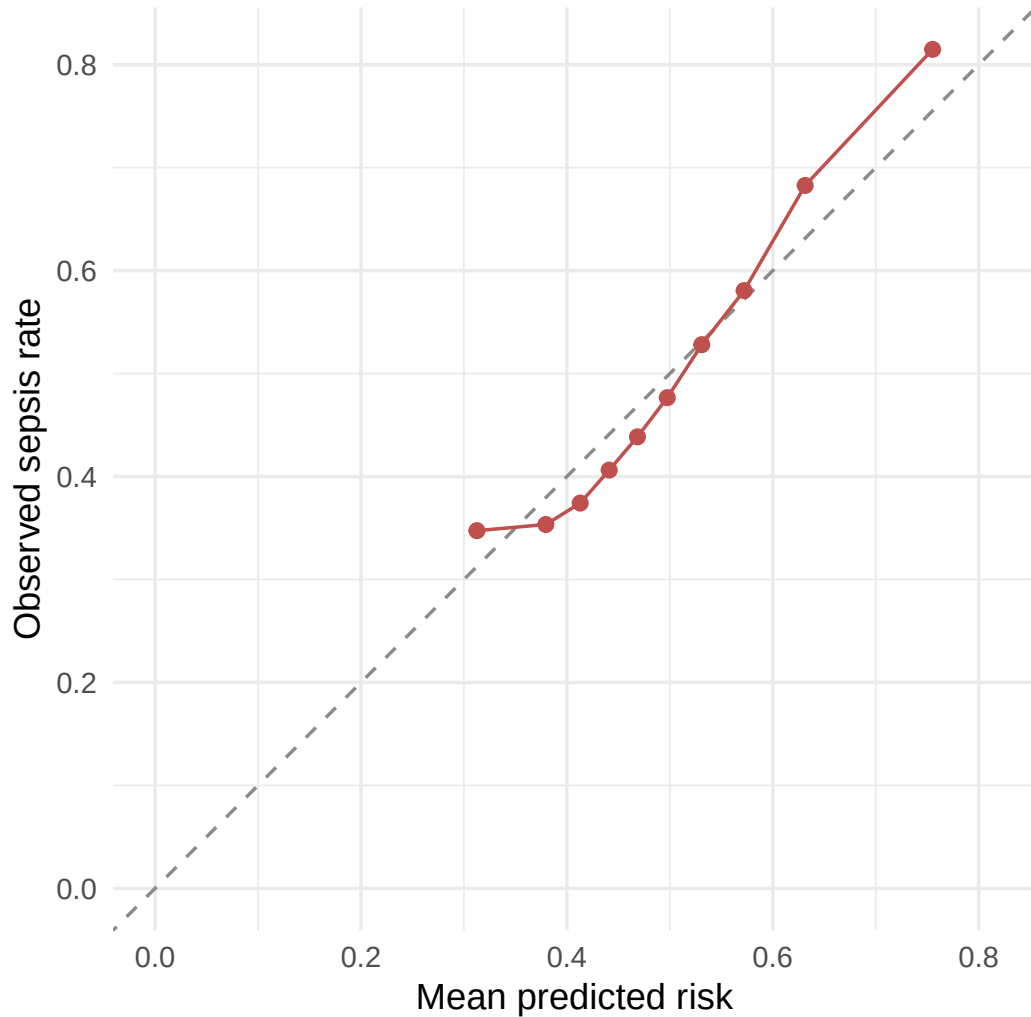


Figure 6: Calibration of the cross-validated six-predictor model (risk deciles). Points on the diagonal indicate predicted risks that match observed frequencies.

The **Brier score** is **0.231** (lower is better), against **0.250** for a null model that always predicts the prevalence — a 8% reduction. The **calibration slope** is **1.02** (1.0 is perfect); Figure 6 shows the decile points lie close to the diagonal, so the predicted probabilities are usable as risks, not merely as a ranking.

Because sepsis is imbalanced and the costs of missing it are asymmetric, ROC-AUC is supplemented with the **precision–recall AUC** and a **decision-curve (net-benefit)** analysis.

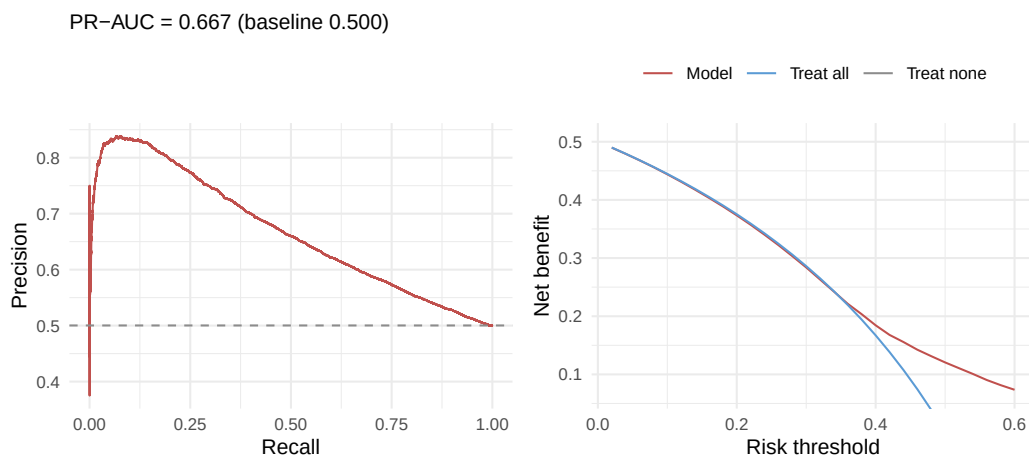


Figure 7: Left: precision-recall curve (dashed line = prevalence, the no-skill baseline). Right: decision curve net benefit of the model against treat-all and treat-none across risk thresholds.

The PR-AUC is **0.667** against a no-skill baseline of **0.500** (the prevalence), and Figure 7 shows the model yields positive net benefit above the treat-all and treat-none strategies across the clinically relevant range of thresholds — the discrimination translates into decision value.

### 8.3 Non-linearity of the lactate effect

The GLM assumed lactate acts **linearly on the log-odds** scale. This is tested by fitting a penalised thin-plate spline `s(lactate_max)` and comparing it with the linear term.

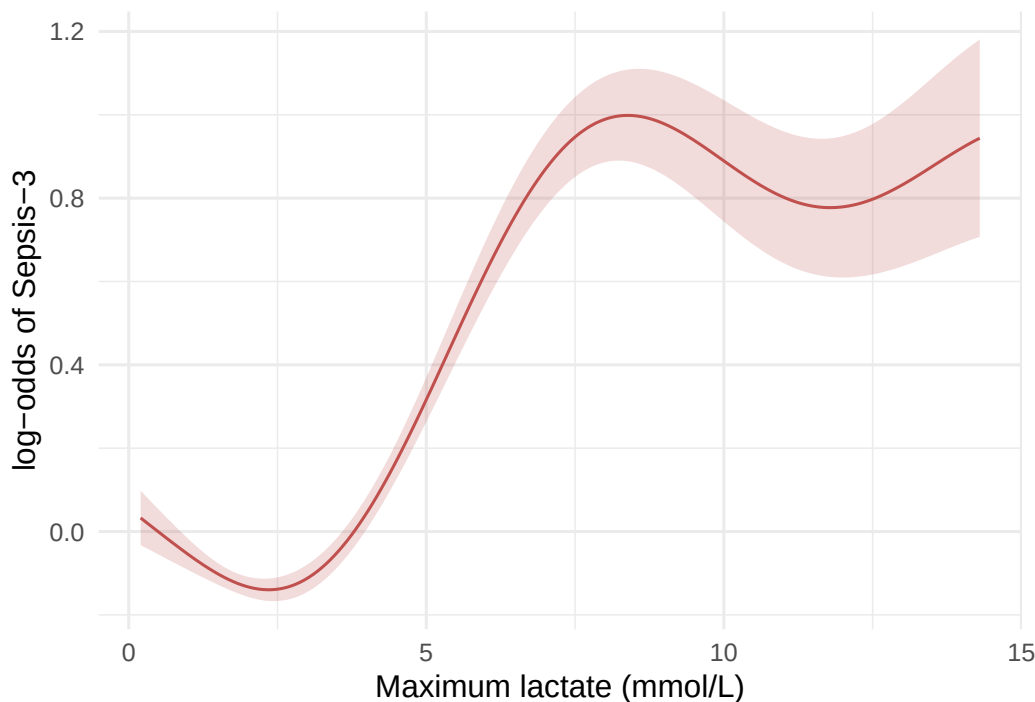


Figure 8: Penalised-spline estimate of the log-odds of sepsis as a function of maximum lactate (shaded 95% band). Curvature indicates a non-linear effect the linear GLM cannot capture.

The spline uses **7.2** effective degrees of freedom (1.0 would mean linear), and it lowers AIC by **223**

relative to the linear term, so the lactate effect is materially non-linear (Figure 8): risk rises steeply at low lactate and then flattens. A production model should therefore spline-transform lactate rather than enter it linearly.

## 8.4 Multicollinearity

Section 5 checked only three correlations. Variance-inflation factors are computed for all six predictors of Model 2.

Table 14: Variance-inflation factors (VIF > 5 signals collinearity).

| Predictor     | VIF  |
|---------------|------|
| hr_max        | 1.14 |
| temp_max_c    | 1.10 |
| lactate_max   | 1.08 |
| age           | 1.07 |
| resp_rate_max | 1.05 |
| wbc_max       | 1.04 |

All VIFs in Table 14 are near 1 (maximum **1.14**), well below the usual cut-off of 5, so the coefficients are not destabilised by collinearity.

## 8.5 The missing-data mechanism

Every inferential section so far used only the 36,529 stays with a **complete** set of the six features — chiefly limited by lactate, which is *ordered at the clinician’s discretion*. Whether a lactate exists is therefore not random: it is itself a marker of concern. This is examined directly.

Lactate is missing in **40.9%** of the full cohort. Sepsis-3 prevalence is **49.8%** among stays with a measured lactate versus **18.7%** among those without (chi-squared  $p < 1e-16$ ) — so *the act of measuring lactate is itself strongly informative*, and complete-case analysis enriches the sample for severity. Missingness here is plausibly **MAR given the recorded vitals** (sicker-looking patients get labs) but a **MNAR** component cannot be excluded.

As a sensitivity analysis the six predictors are **multiply imputed** (MICE, predictive mean matching) on the full cohort and the logistic model re-fitted, pooling coefficients by Rubin’s rules.

Table 15: Odds ratios: complete-case versus multiply-imputed full cohort.

|   | Predictor     | OR (complete-case) | OR (imputed cohort) | 95% CI (imputed) |
|---|---------------|--------------------|---------------------|------------------|
| 2 | lactate_max   | 1.081              | 1.089               | [1.079, 1.100]   |
| 3 | wbc_max       | 1.018              | 1.034               | [1.031, 1.038]   |
| 4 | hr_max        | 1.008              | 1.011               | [1.010, 1.012]   |
| 5 | resp_rate_max | 1.003              | 1.000               | [1.000, 1.000]   |
| 6 | temp_max_c    | 1.685              | 1.954               | [1.902, 2.007]   |
| 7 | age           | 1.011              | 1.013               | [1.012, 1.014]   |

Table 15 shows the odds ratios keep the same direction and comparable magnitude after imputing the

full cohort: the associations are not an artefact of the complete-case restriction, though the absolute sepsis prevalence differs between the samples. This missingness signal is itself a lesson for the real-time model — *whether and when a test is ordered* is a usable predictor, not just a nuisance.

*Collectively, this section shows the six-predictor model generalises out of sample, is well calibrated and clinically useful, benefits from a non-linear lactate term, is free of collinearity, and is robust to the missing-data mechanism — while flagging that measurement itself carries information.*

## 9 Conclusion

This report asked whether routine physiology that is *not* part of the sepsis definition can identify Sepsis-3, and what different statistical methods reveal about the same data.

- **Hypothesis tests** (Section 3) found genuine differences between septic and non-septic stays, largest for lactate (Cohen’s  $d = -0.20$ ), and weaker but real for white-cell count and tachypnoea.
- The **GLM** (Section 5) showed lactate is the single strongest predictor, while vital signs and age add independent information: AIC fell from 50257 to 48112, with an approximate AUC of 0.66.
- The **mixture model** (Section 6.2) and **hidden Markov model** (Section 7.2) both recovered two lactate populations without labels, agreeing almost exactly with each other (54.3% vs 53.8%). Crucially, lactate alone only partially recovers sepsis — a reminder that the diagnosis is multi-dimensional.
- The **MLE** (Section 4) and **MCMC** (Section 7.1) agreed closely on the typical septic lactate ( $\approx 2.37$  mmol/L), with nearly identical interval estimates.
- **Validation** (Section 8) confirmed the six-predictor model holds up out of sample (10-fold CV AUC 0.661, DeLong  $p < 1e-16$  versus lactate alone), is well calibrated (Brier 0.231, slope 1.02), gains from a non-linear lactate term, and keeps its associations under multiple imputation — while showing that *whether* a lactate was ordered is itself predictive.

*The methodological message echoes across tools: several independent methods converge on the same, clinically coherent picture. Unlike a near-separable marker such as tumour radius, no single physiological variable identifies sepsis on its own — which is precisely why Sepsis-3 couples suspected infection with a multi-organ SOFA score. The geometry of one biomarker is not enough; it takes a combination of signals to recognise a dysregulated response to infection.*

**Limitations and the path to real-time prediction.** The analysis subset is restricted to stays with a measured lactate, which enriches for severity (quantified and partly addressed by the multiple-imputation sensitivity analysis in Section 8.5); sepsis was operationalised with a first-24h (“day-1”) SOFA and a small set of documented simplifications (Section 05 header); and the ordering-based HMM is illustrative, since stays are not a true time series. Most importantly, this report is **cross-sectional**: it collapses each stay into one row of first-24h summaries and answers “*was this stay septic?*” rather than the project’s ultimate question — “*will this patient become septic in the next few hours, given the data so far?*” That real-time question requires an **hourly patient panel**, a defined **sepsis-onset time** and prediction horizon, **survival / time-varying-hazard** modelling of time to onset, a **within-patient** (not ordering-based) HMM, and **time-dependent, leakage-safe** evaluation benchmarked against streaming clinical scores (qSOFA, SIRS, hourly SOFA, MEWS). Building that longitudinal foundation is the subject of the follow-on work, for which the associations, effect sizes, non-linearities, and missingness structure established here provide the modelling priors.

## Acknowledgment

During this project, I applied the **PAIR framework** to responsibly use generative AI alongside primary sources.

- **Problem:** Build a reproducible MIMIC-IV Sepsis-3 pipeline and a layered statistical analysis (EDA, hypothesis testing, MLE, GLMs, density and mixture modelling, Metropolis-Hastings sampling, and a hidden Markov model) in R, supported by a coherent written report.
- **AI:** I used AI assistants to discuss implementation choices, clarify statistical concepts, and refine the writing.
- **Interaction:** All AI-generated suggestions were critically reviewed, checked against primary sources (Singer et al., 2016; Seymour et al., 2016; Vincent et al., 1996; Silverman, 1986; Robert & Casella, 2004; Gelman et al., 2013; Rabiner, 1989; Visser & Speekenbrink, 2010), and adapted to the structure and seed-based reproducibility of this work.
- **Reflection:** AI assistance accelerated drafting and improved clarity, while the interpretation, code, model choices, and conclusions remain my own.

## References

1. Singer, M., Deutschman, C. S., Seymour, C. W., et al. (2016). The Third International Consensus Definitions for Sepsis and Septic Shock (Sepsis-3). *JAMA*, 315(8), 801–810.
2. Seymour, C. W., Liu, V. X., Iwashyna, T. J., et al. (2016). Assessment of Clinical Criteria for Sepsis (Sepsis-3). *JAMA*, 315(8), 762–774.
3. Vincent, J. L., Moreno, R., Takala, J., et al. (1996). The SOFA (Sepsis-related Organ Failure Assessment) score to describe organ dysfunction/failure. *Intensive Care Medicine*, 22(7), 707–710.
4. Johnson, A. E. W., Bulgarelli, L., Shen, L., et al. (2023). MIMIC-IV, a freely accessible electronic health record dataset. *Scientific Data*, 10, 1.
5. Cohen, J. (1988). *Statistical Power Analysis for the Behavioral Sciences* (2nd ed.). Lawrence Erlbaum Associates.
6. Silverman, B. W. (1986). *Density Estimation for Statistics and Data Analysis*. Chapman and Hall.
7. Dempster, A. P., Laird, N. M., & Rubin, D. B. (1977). Maximum likelihood from incomplete data via the EM algorithm. *JRSS B*, 39(1), 1–38.
8. Gelman, A., Carlin, J. B., Stern, H. S., et al. (2013). *Bayesian Data Analysis* (3rd ed.). CRC Press.
9. Robert, C., & Casella, G. (2004). *Monte Carlo Statistical Methods* (2nd ed.). Springer.
10. Rabiner, L. R. (1989). A tutorial on hidden Markov models and selected applications in speech recognition. *Proceedings of the IEEE*, 77(2), 257–286.
11. Visser, I., & Speekenbrink, M. (2010). depmixS4: An R package for hidden Markov models. *Journal of Statistical Software*, 36(7), 1–21. ““